

Lab Program 9

Title: Organise Kubernetes Pods using Labels

File Structure:

prog9/

namespaces.yaml

pods.yaml

STEP 1: Start Minikube

\$ minikube start

```
victus% minikube start
🐳 minikube v1.37.0 on Ubuntu 24.04
🌟 Using the docker driver based on existing profile
👍 Starting "minikube" primary control-plane node in "minikube" cluster
🚧 Pulling base image v0.0.48 ...
🔄 Restarting existing docker container for "minikube" ...
```

STEP 2: Create namespaces.yaml

\$ nano namespaces.yaml

```
apiVersion: v1
kind: Namespace
metadata:
  name: dev
---
apiVersion: v1
kind: Namespace
metadata:
  name: prod
```

STEP 3: Create pods.yaml

\$ nano pods.yaml

```
apiVersion: v1
kind: Pod
metadata:
  name: frontend-pod
  namespace: dev
  labels:
    app: frontend
    env: dev
spec:
  containers:
    - name: frontend-container
      image: nginx:latest
      ports:
        - containerPort: 80
```

```
apiVersion: v1
kind: Pod
metadata:
  name: backend-pod
  namespace: dev
  labels:
    app: backend
    env: dev
spec:
  containers:
    - name: backend-container
      image: nginx:latest
      ports:
        - containerPort: 80
```

```
apiVersion: v1
kind: Pod
metadata:
  name: frontend-pod
  namespace: prod
  labels:
    app: frontend
    env: prod
spec:
  containers:
    - name: frontend-container
      image: nginx:latest
      ports:
        - containerPort: 80
```

```
apiVersion: v1
kind: Pod
metadata:
  name: backend-pod
  namespace: prod
  labels:
    app: backend
    env: prod
spec:
  containers:
    - name: backend-container
      image: nginx:latest
      ports:
        - containerPort: 80
```

STEP 4: Apply the namespaces.yaml and then pods.yaml

\$ kubectl apply -f namespaces.yaml

```
victus% kubectl apply -f namespaces.yaml
namespace/dev created
namespace/prod created
```

\$ kubectl apply -f pods.yaml

```
victus% kubectl apply -f pods.yaml
pod/frontend-pod created
pod/backend-pod created
pod/frontend-pod created
pod/backend-pod created
```

STEP 5: Verify pods of both namespaces

\$ kubectl get pods -n dev

```
victus% kubectl get pods -n dev
NAME                READY   STATUS    RESTARTS   AGE
backend-pod         1/1     Running   0           19s
frontend-pod        1/1     Running   0           19s
```

\$ kubectl get pods -n prod

```
victus% kubectl get pods -n prod
NAME                READY   STATUS    RESTARTS   AGE
backend-pod         1/1     Running   0           31s
frontend-pod        1/1     Running   0           31s
```

STEP 6: Filter pods by Label

\$ kubectl get pods -n dev -l app=frontend

```
victus% kubectl get pods -n dev -l app=frontend
NAME                READY   STATUS    RESTARTS   AGE
frontend-pod        1/1     Running   0           53s
```

\$ kubectl get pods -n prod -l app=backend

```
victus% kubectl get pods -n prod -l app=backend
NAME                READY   STATUS    RESTARTS   AGE
backend-pod         1/1     Running   1 (44m ago)  51m
```